CCSE R82 Lab Notes and Corrections

Please use these notes as you work through the lab in the Kortext book

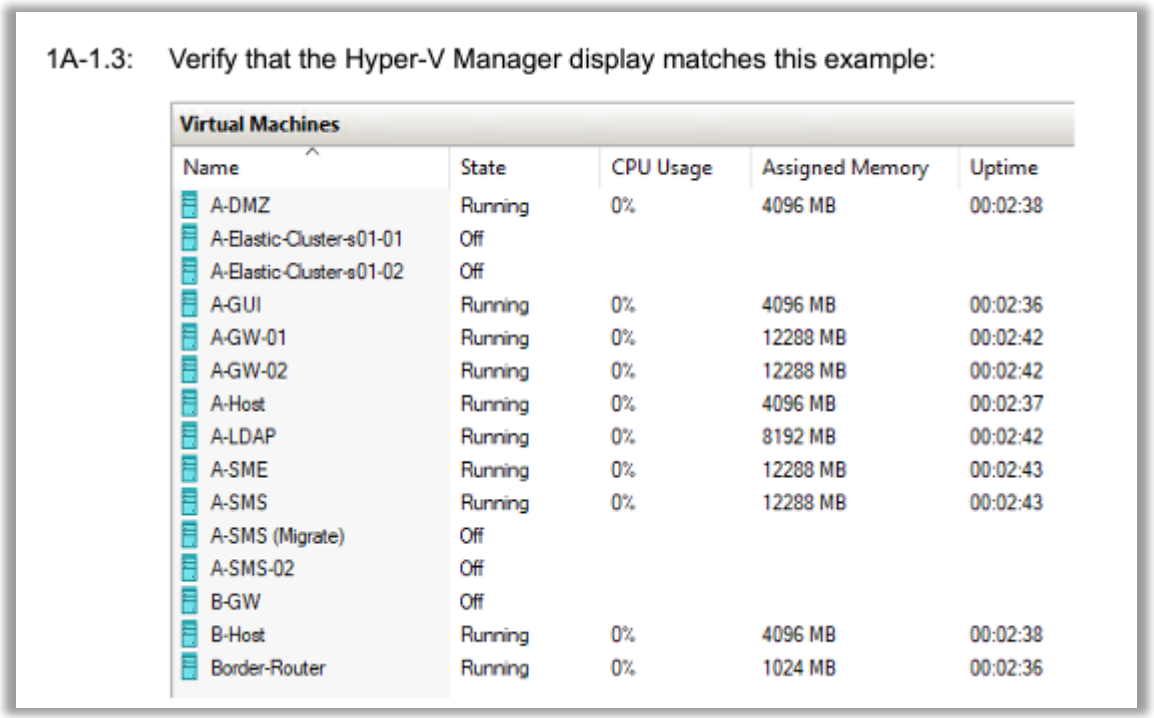
Hyper-V Manager

Please ignore the references to Hyper-V Manager in all labs.

All the VMs that are used in the first 5 labs are booted up when you Launch the lab.

Lab 6 & 7: Booting other VMs during the migration and ElasticXL lab is easy to do in the Skillable platform interface/portal.

There is no Hyper-V Manager in the Skillable labs. This is not a problem.



1A-1.3: Verify that the Hyper-V Manager display matches this example:

Virtual Machines

Name	State	CPU Usage	Assigned Memory	Uptime
A-DMZ	Running	0%	4096 MB	00:02:38
A-Elastic-Cluster-s01-01	Off			
A-Elastic-Cluster-s01-02	Off			
A-GUI	Running	0%	4096 MB	00:02:36
A-GW-01	Running	0%	12288 MB	00:02:42
A-GW-02	Running	0%	12288 MB	00:02:42
A-Host	Running	0%	4096 MB	00:02:37
A-LDAP	Running	0%	8192 MB	00:02:42
A-SME	Running	0%	12288 MB	00:02:43
A-SMS	Running	0%	12288 MB	00:02:43
A-SMS (Migrate)	Off			
A-SMS-02	Off			
B-GW	Off			
B-Host	Running	0%	4096 MB	00:02:38
Border-Router	Running	0%	1024 MB	00:02:36

Lab 1

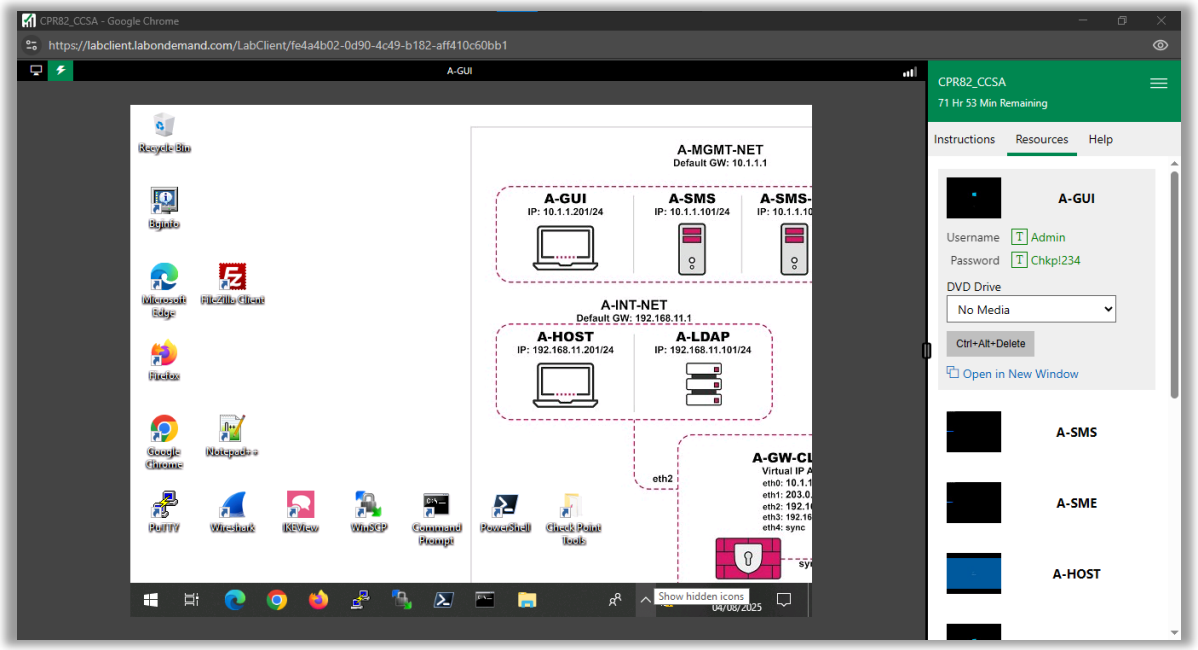
Lab start page: 39

Lab end page: 81

You do not need to Start any VMs in your lab.

All VMs are started and booted up when the Skillable lab is Launched

Skillable Lab portal:

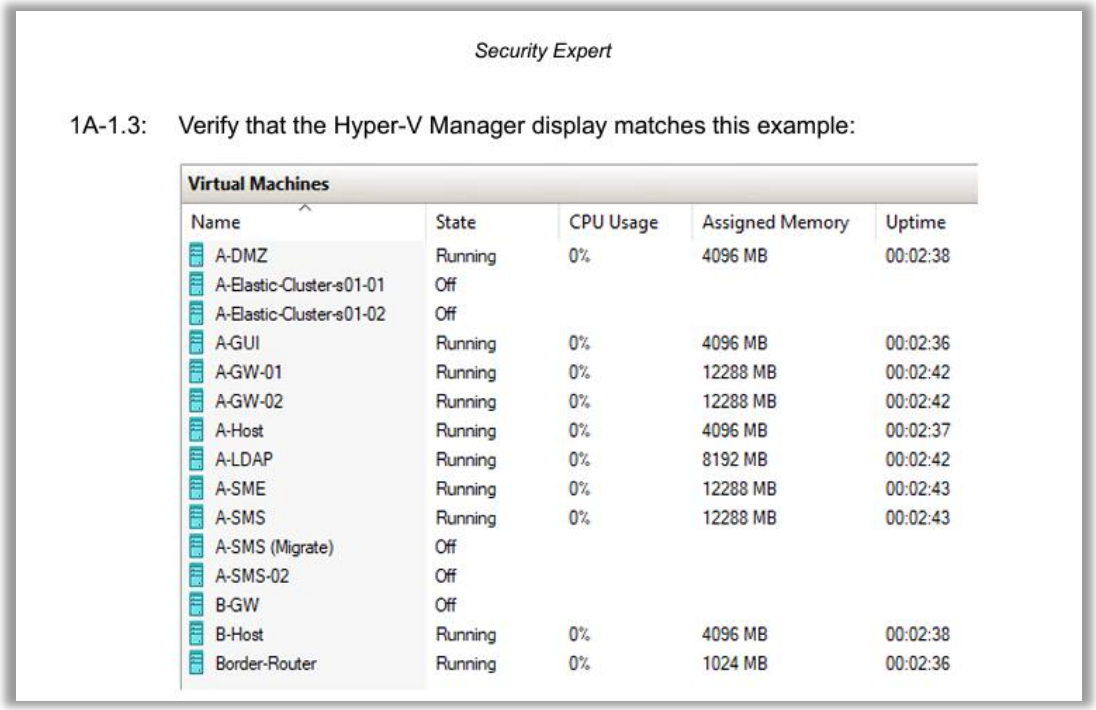


Ignore the reference to Hyper-V Manager in this lab and in all lab tasks and lab steps.

All the VMs that are needed for the first 5 Labs are booted up when you Launch the lab.

Page 44 – Step 1A – 1.3

Ignore the reference to Hyper-V Manager on page 44.



Security Expert

1A-1.3: Verify that the Hyper-V Manager display matches this example:

Virtual Machines

Name	State	CPU Usage	Assigned Memory	Uptime
A-DMZ	Running	0%	4096 MB	00:02:38
A-Elastic-Cluster-s01-01	Off			
A-Elastic-Cluster-s01-02	Off			
A-GUI	Running	0%	4096 MB	00:02:36
A-GW-01	Running	0%	12288 MB	00:02:42
A-GW-02	Running	0%	12288 MB	00:02:42
A-Host	Running	0%	4096 MB	00:02:37
A-LDAP	Running	0%	8192 MB	00:02:42
A-SME	Running	0%	12288 MB	00:02:43
A-SMS	Running	0%	12288 MB	00:02:43
A-SMS (Migrate)	Off			
A-SMS-02	Off			
B-GW	Off			
B-Host	Running	0%	4096 MB	00:02:38
Border-Router	Running	0%	1024 MB	00:02:36

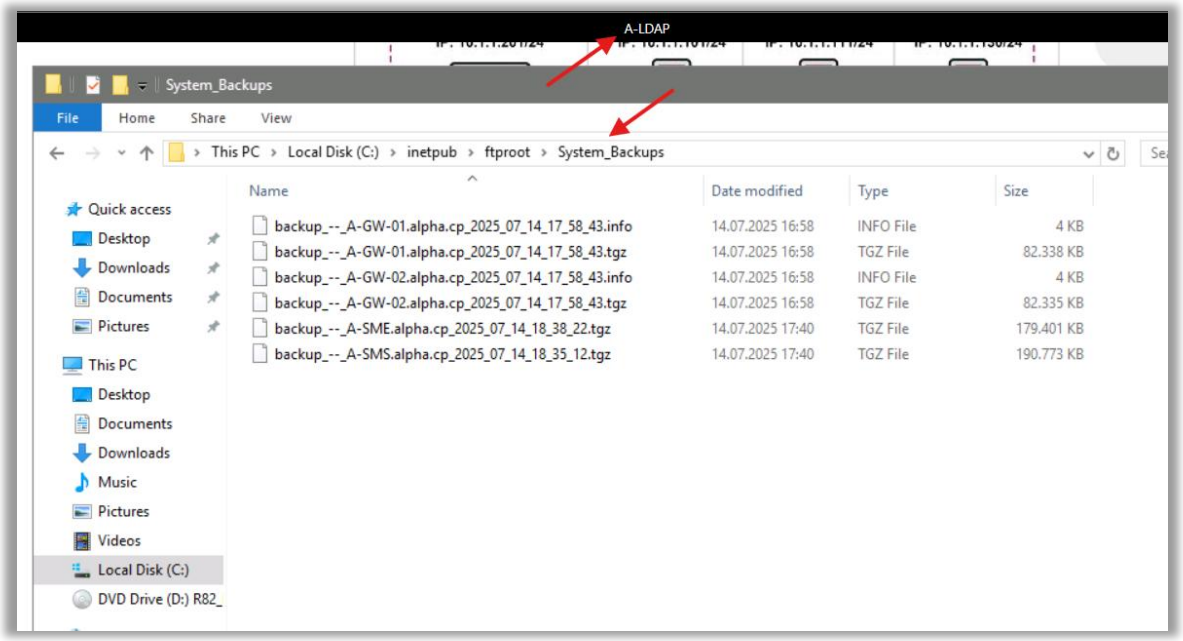
Page 47:

1A - 1.13

The screenshot is wrong.

The upload path is correct in the instructions (/System_Backup**s/**)

That is a folder on the A-LDAP VM/Server:



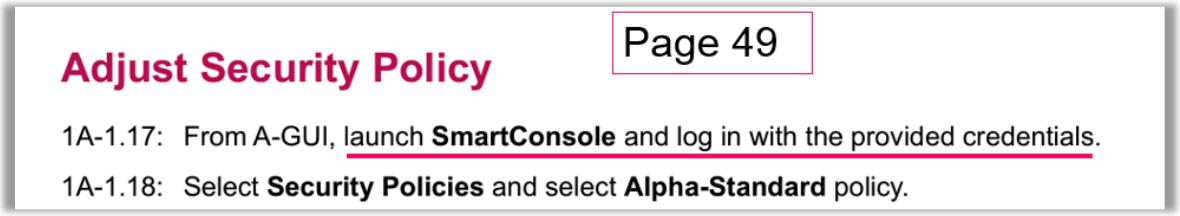
Page 49:

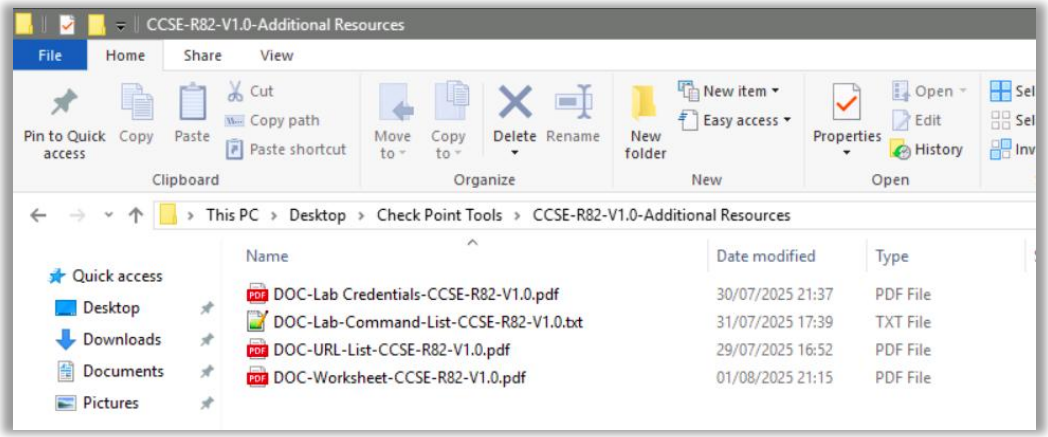
For all lab credentials please refer to Page 41 & 42

For SmartConsole: cpadmin

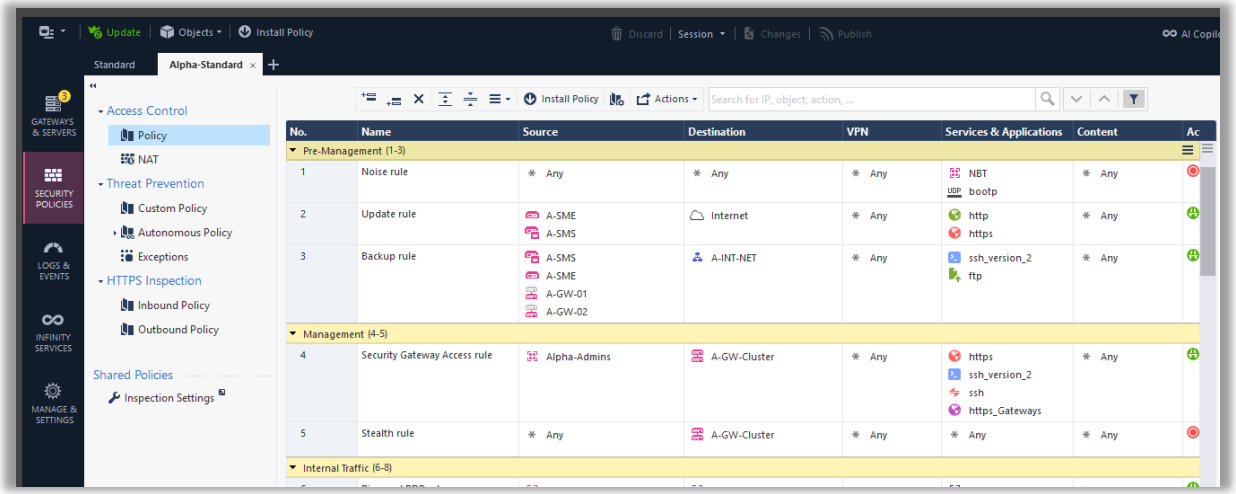
The credentials are also stored in a PDF in **A-GUI**:

Desktop > **Check Point Tools** > **CCSE-R82-V1.0-Additional-Resources**





Make sure you open the **Alpha-Standard** policy package.

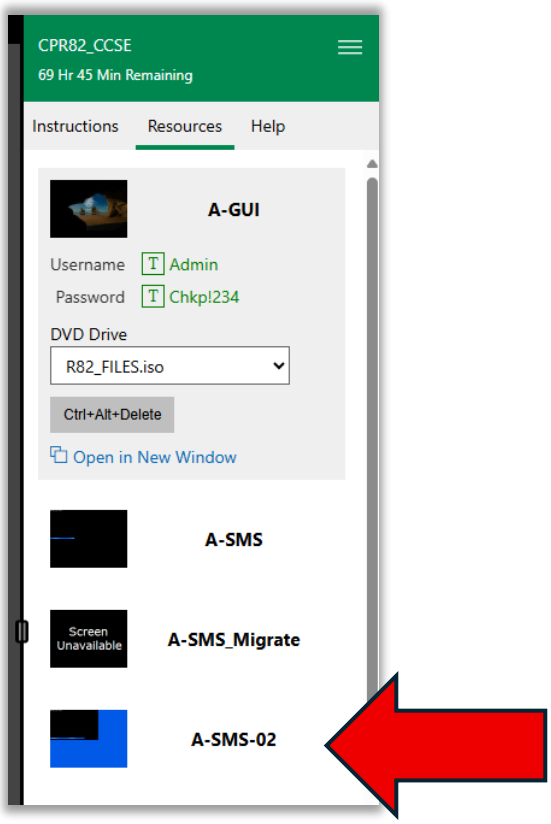


Page 51:

Ignore **Step 1A - 2.1, move to Step 1A – 2.2** (see screenshots below)

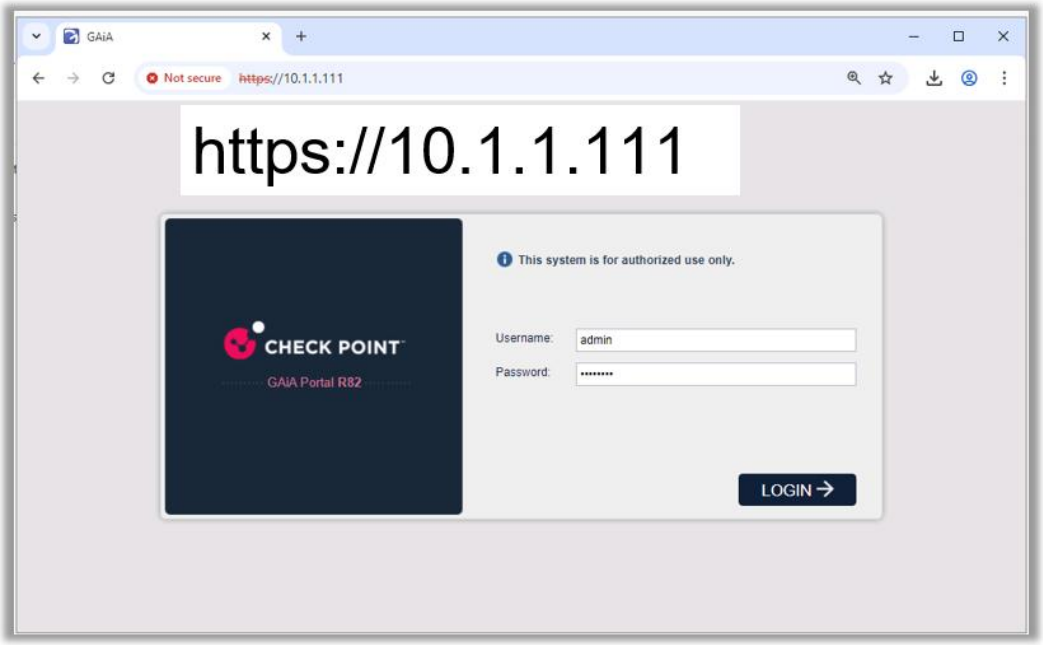
A-SMS-02 is already booted up.

We do not have the Hyper-V Manager GUI in the Skillable lab.



1A-2.2

The Gaia Portal for A-SMS-02 is accessible via https://10.1.1.111



Page 52 1A-2.7:

NTP - Please use your preferred time zone and not Chicago, America (GMT-6:00.

Page 56, 57 & 58:

- Ignore the note on Page 56 Step 1A – 2.17 regarding hotfix versions
- Skip page 57 & 58

No Jumbo Hot Fixes needed for Lab 1

Please ignore reference to JHFA 33

Optional:

You can install JHFA 91 from the locally stored file, but it will take around 45 minutes.

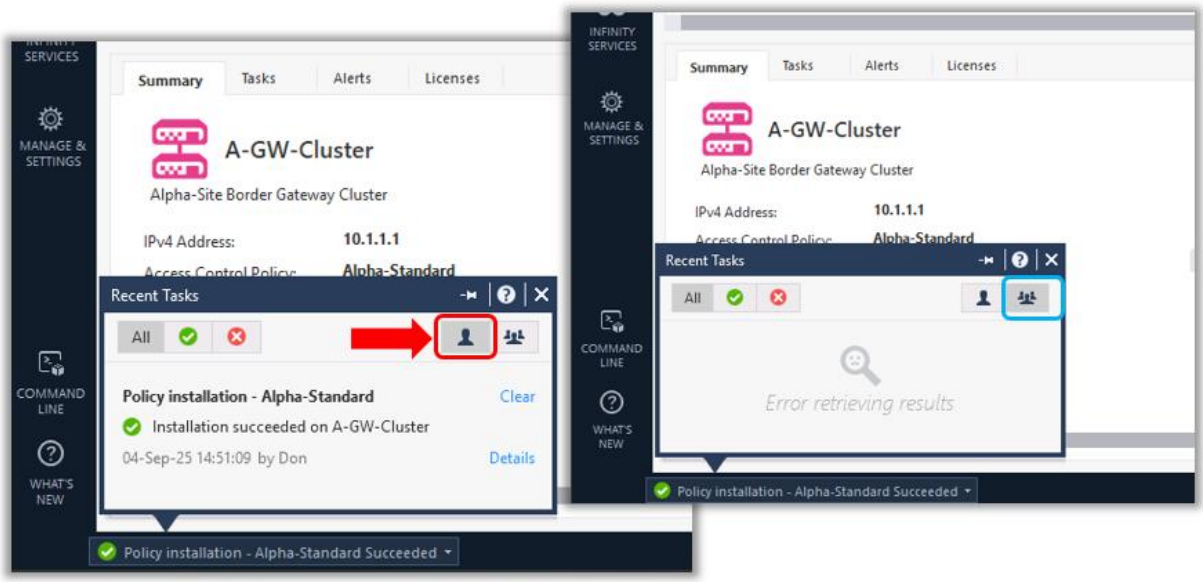
A-GUI > Desktop > Check Point Tools:

Check_Point_R82_jumbo_hf_main_Bundle_T91_FULL.tar

If you install JHFA 91 to A-SMS then you must also install it onto A-SMS-02.

Page 62:

If you see an Error in the Recent Tasks window, then click the **My tasks** button as shown by the red arrow below**.**



Lab 2

Lab start page: 127

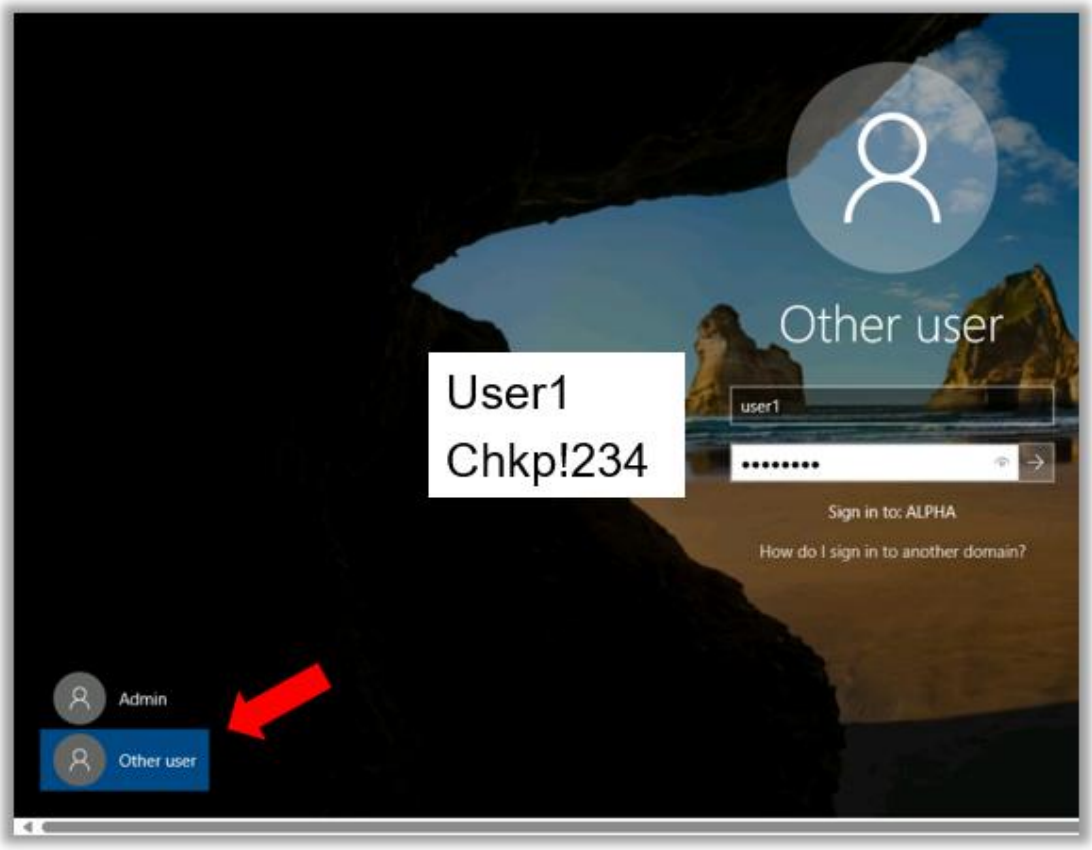
Lab end page: 183

Page 135:

To log into **A-Host** as **User1** you must log out first (Admin is logged in).

On A-Host:

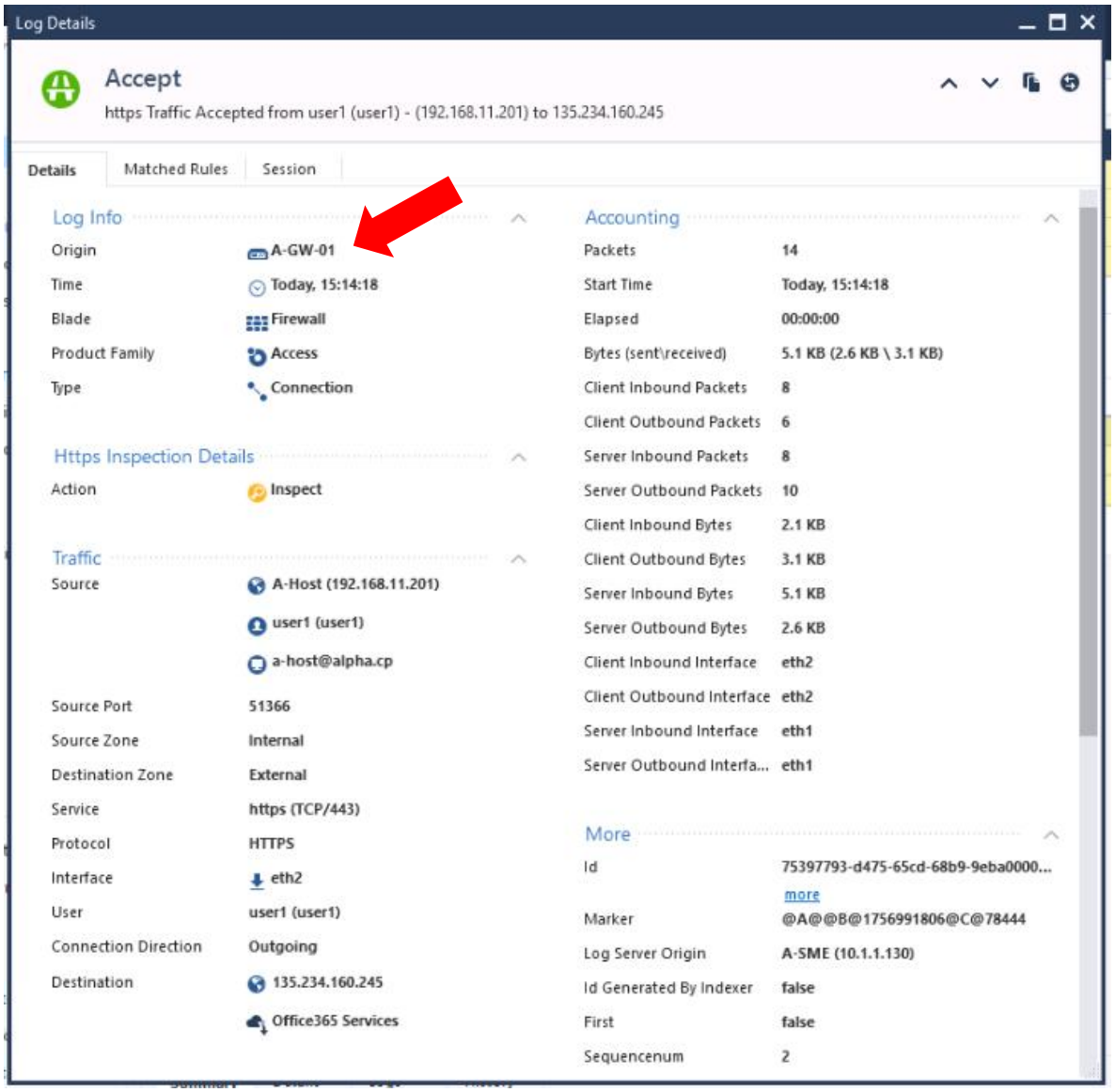
- **Right click**  **Start** > Click **Sign out**
- **Click Other user** (in the bottom left-hand corner of the Windows 10 login screen).



Page 136 & 137:

The screenshots show ElasticXL instead of A-GW-01 (or 02)

The log screenshot should look like this



Lab 2B Notes

The Story at the start of the lab (Page 139) is interesting.

Lab

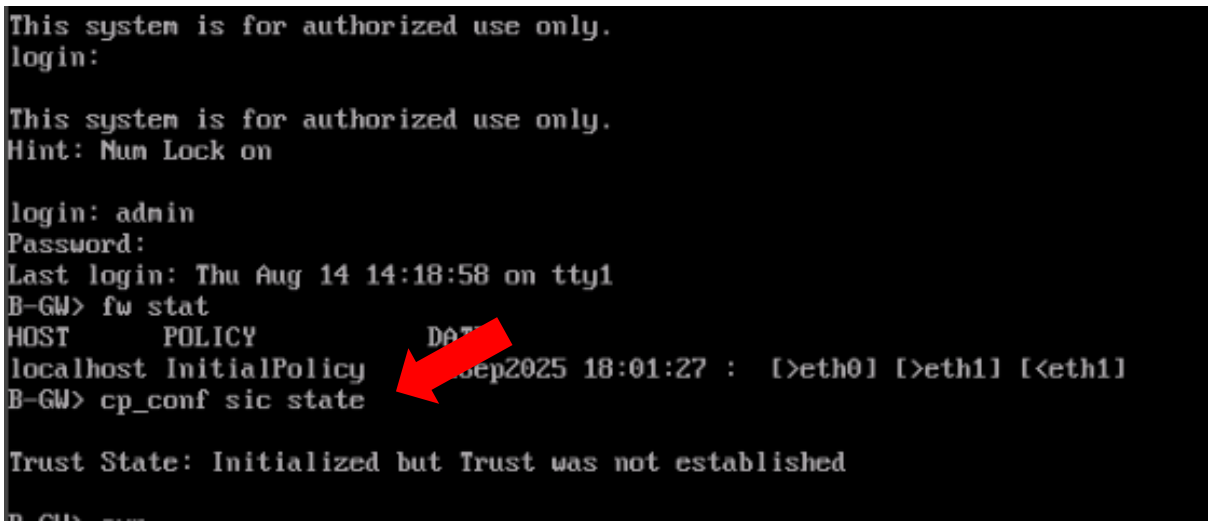
2B

Configure Network Address Translation Policy

As part of the acceptance testing for the new security environment, the Web Development team has been testing their access to the A-DMZ web server. Everything was going fine until the new web developer tried to connect to the server using the external address and nothing happened. The new developer is at Bravo Site while the majority of the development team is at headquarters. Web development has requested that the Security Team look into the issue.

Yesterday, the Primary Security Management Server at Bravo-Site failed without any current backups. The CISO has decided that recovery is not cost effective because the Bravo-Site was preparing to move to the Alpha Corporation management in the next week anyway. The decision to move forward with moving the B-GW to A-SMS management was handed over this morning and your team lead has assigned you the tasks to resolve these issues.

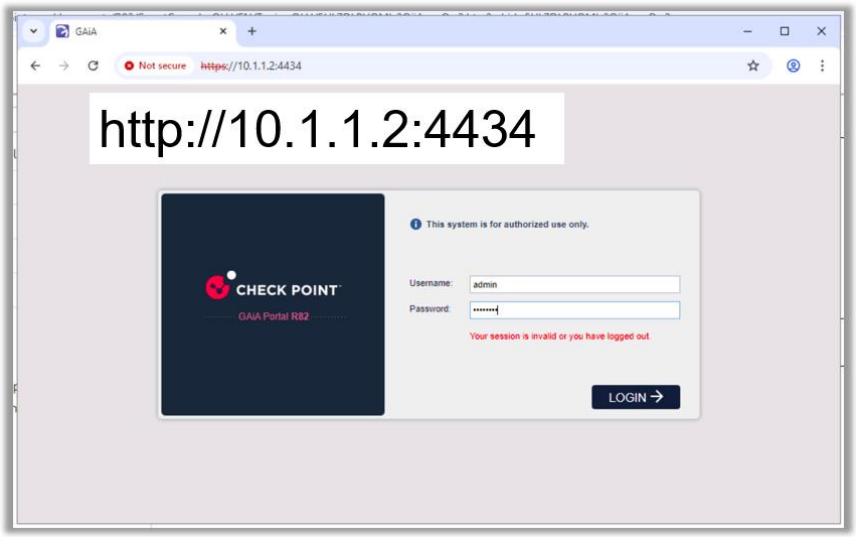
The B-GW is built but never had a SIC trust with any server.



Page 155:

Step 2B-3.8

The Gaia Portals for the gateways use port 4434



Page 155:

Proxy ARP configuration using CLISH (an alternative to the Gaia Portal)

A-GW-01 (in CLISH)

add proxy-arp ip 203.0.113.121 real-ip 203.0.113.2 interface eth1

save config

A-GW-02 (in CLISH)

add proxy-arp ip 203.0.113.121 real-ip 203.0.113.3 interface eth1

save config

```
A-GW-01>
A-GW-01> add arp proxy ipv4-address 203.0.113.121 interface eth1 real-ipv4-address 203.0.113.2
A-GW-01> save config
A-GW-01>
A-GW-01>
A-GW-01> show arp proxy all
IP Address              MAC Address / Interface         Real IP Address
203.0.113.121           eth1                            203.0.113.2

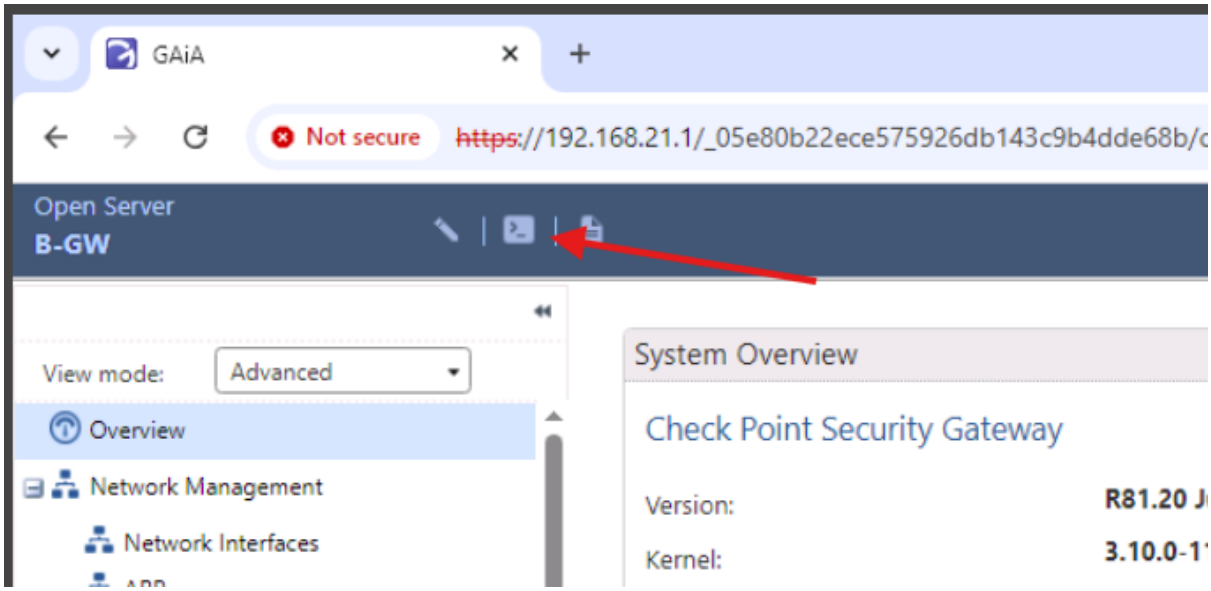
A-GW-01>
```

Page 159:

Step 2B-4.4

2B-4.4: Click **Terminal** and log in to B-GW with the provided credentials.

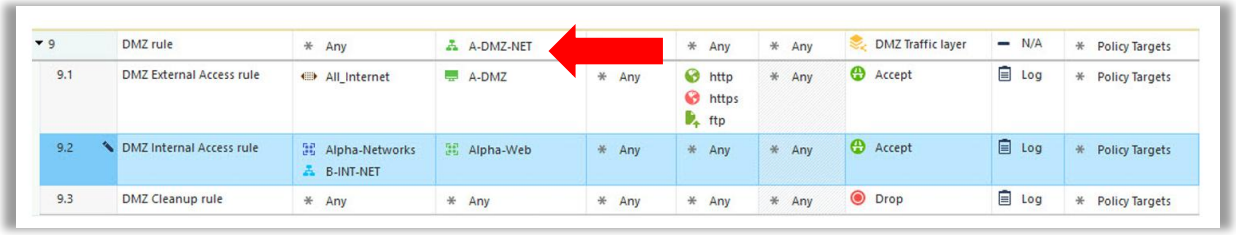
The Terminal button is in the top left side of the Gaia Portal.



Page 172, 2B-4.50:

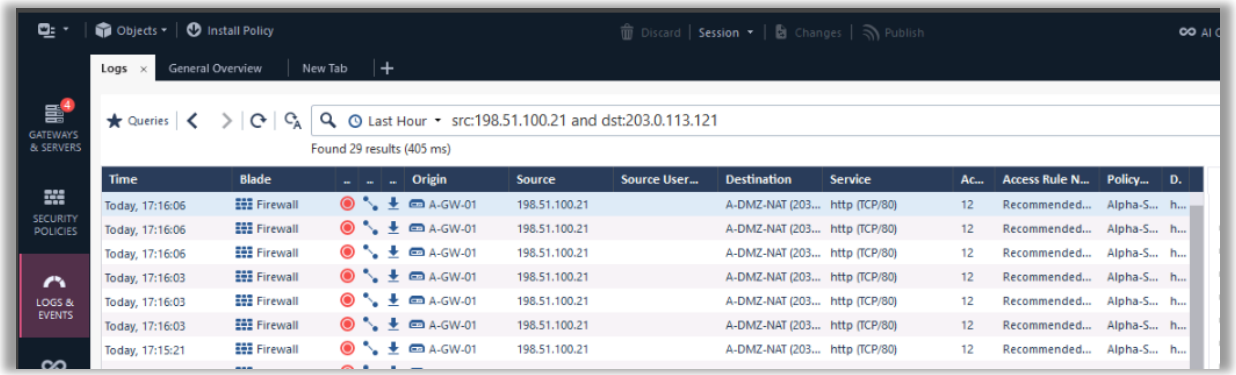
Rule 9 should only include **A-DMZ-NET**

Don’t make the mistake of adding A-DMZ-NAT



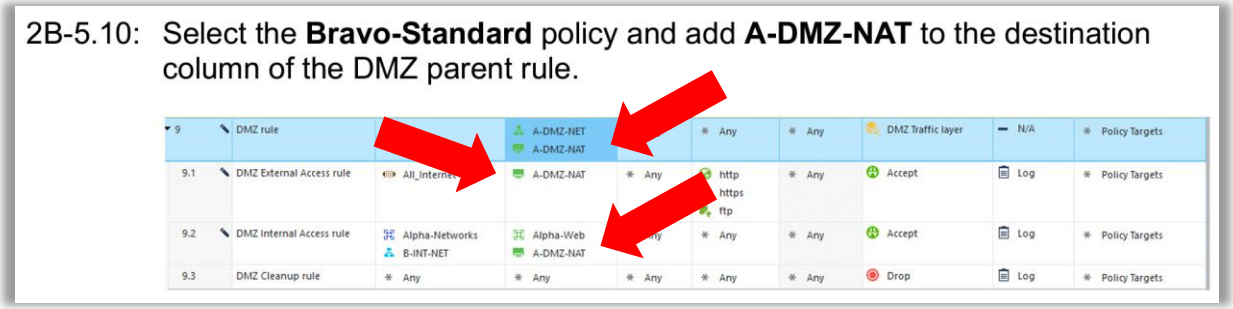
Page 178

src:198.51.100.21 and dst:203.0.113.121



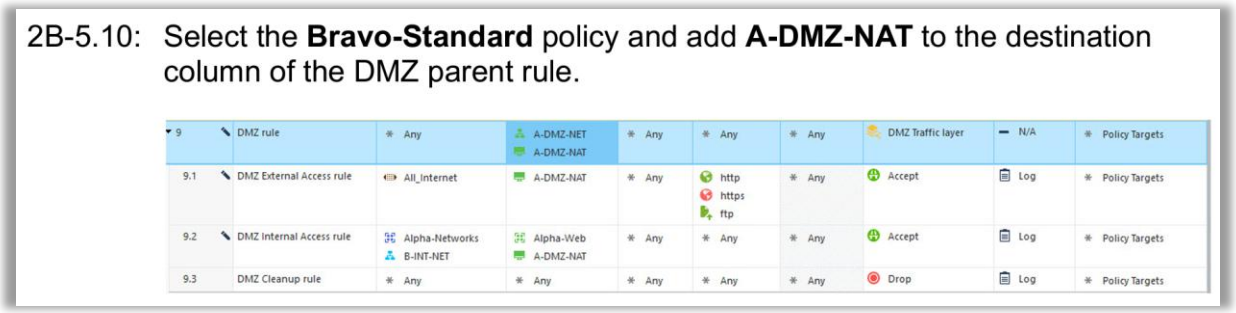
Page 179

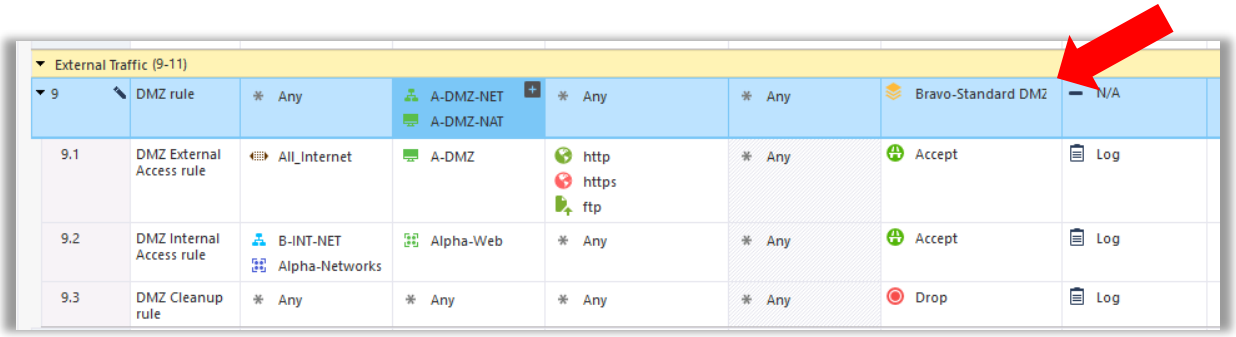
Please make sure that you add the A-DMZ-NAT object to the parent and first two sub rules



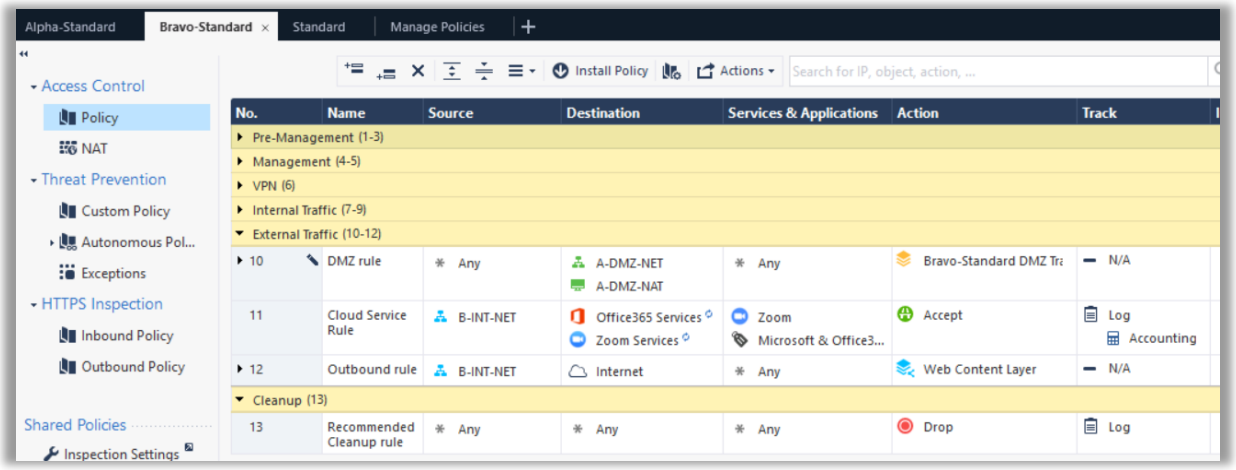
Page 179:

The screenshot in the book is for Alpha and not Bravo. Mistake in the book.



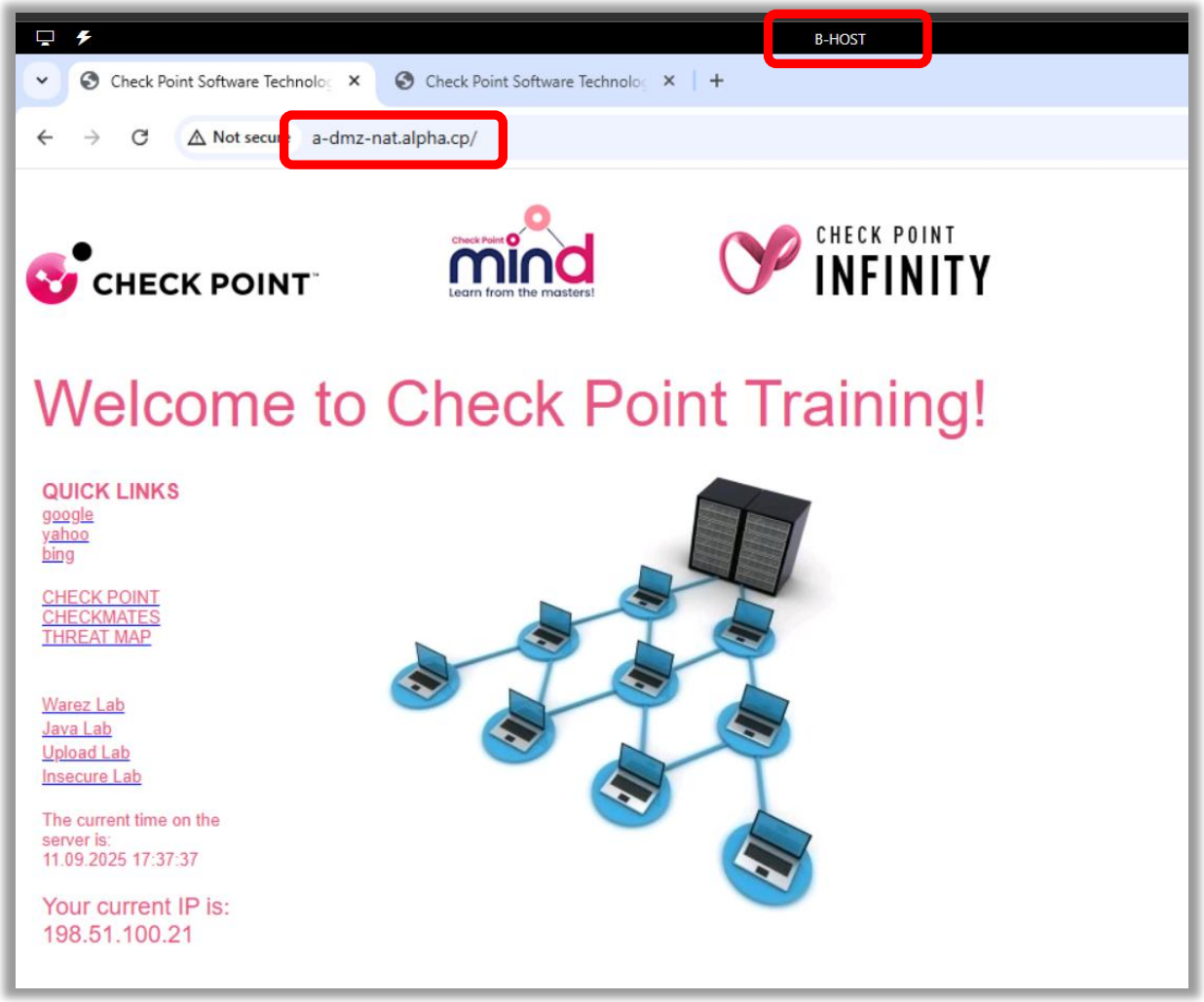


Bravo-Standard > **Policy** > **Rule 10**



Page 182:

Successful access to A-DMZ from B-Host



Lab 3

Lab start page: 203

Lab end page: 234

No lab notes for Lab 3

Lab 4

Lab start page: 259

Lab end page: 341

Page 332:

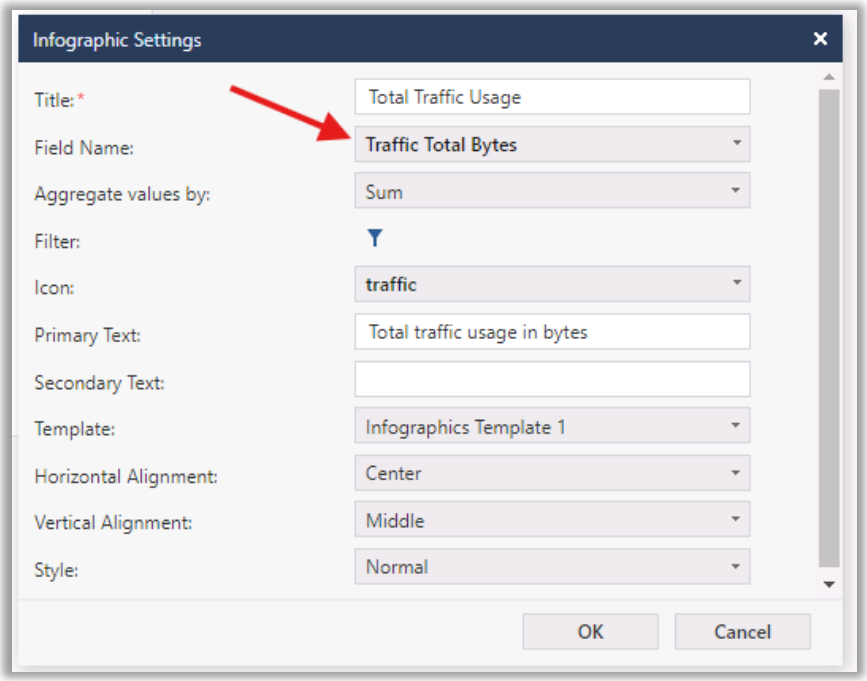
If **FW105 is blank** after making changes, then please do a Rescan:

MANAGE & SETTINGS > **Blades** > **Compliance** > **Settings** > **Rescan** button

Page 309:

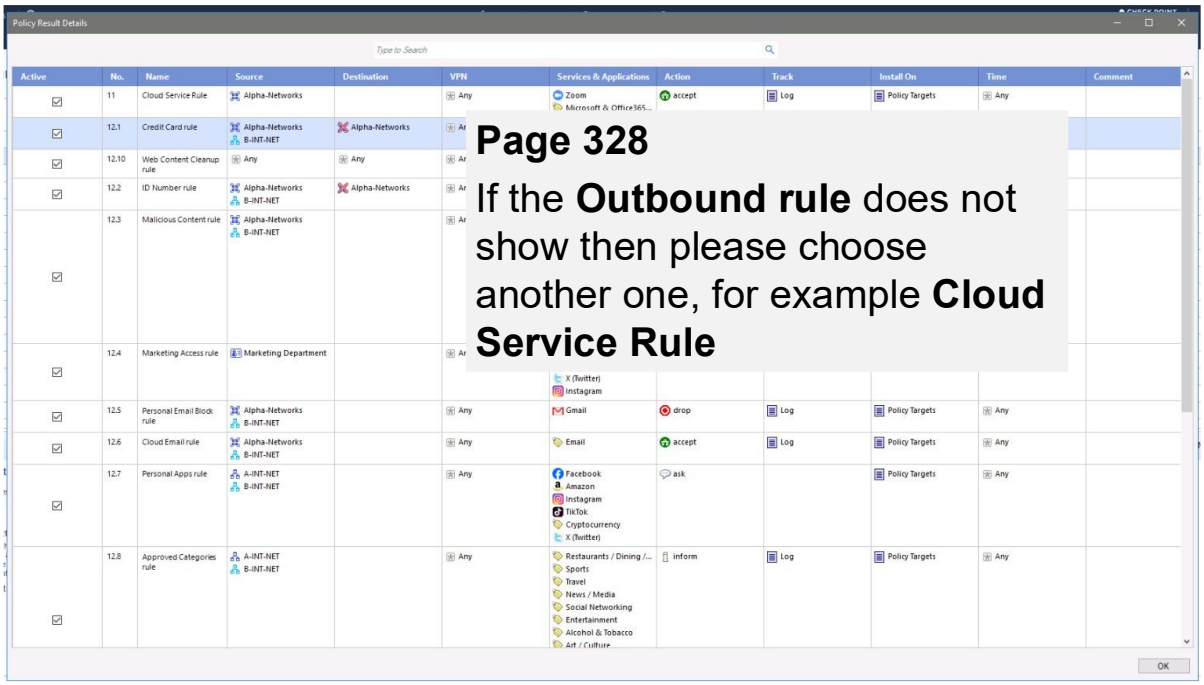
4A – 5.37

Please select **Traffic Total Bytes** in the **Field Name** cell**.**



Page 328:

If the **Outbound rule** does not show then please choose another one, for example **Cloud Service Rule**



Lab 5

Lab start page: 365

Lab end page: 385

Page 370:

The upload of the **R82 upgrade file** (5GB tar file) will take **10 minutes** or more.

You can take a coffee break at that time.

You can also do the **B-GW backup** steps on **Page 373** while the tar file uploads to the repository.

Page 374:

The upgrade takes about **40 minutes**.

- Task 5A-2: Install Jumbo Hotfixes with Central Deployment

Page 370:

Please use the Take 779 package (tar):

Check_Point_R82_T779_Gaia_Install_and_Upgrade.tar

The file is stored in → A-GUI > Desktop > Check Point Tools:

Check_Point_R82_T779_Gaia_Install_and_Upgrade.tar

The upload of the **R82 upgrade file** (5GB tar file) can take **10 minutes** or more.

- You can take a break at that time, or, if the A-GUI is responsive enough;
- You can start to do the **B-GW backup** steps on **Page 373** while the tar file uploads to the repository.

Page 374:

The upgrade takes about **40 minutes**.

Page 377:

Please use the Take 91 file. The Take 33 file has been deleted from the lab.

A-GUI > Desktop > Check Point Tools

Check_Point_R82_jumbo_hf_main_Bundle_T91_FULL.tar

The **upload** of the JHFA file and the deployment will take time.

The JHFA Take 91 file is **2.3GB**

Estimated time: **45 minutes**

Page 380:

If the JHFA 91 does not appear in the search bar, then paste the filename into the search bar.

Check_Point_R82_jumbo_hf_main_Bundle_T91_FULL.tar

To easily copy the file name in the lab, browse to:

Desktop > Check Point Tools >

Check_Point_R82_jumbo_hf_main_Bundle_T91_FULL.tar

Note:

In some cases we must use the file extension .tgz

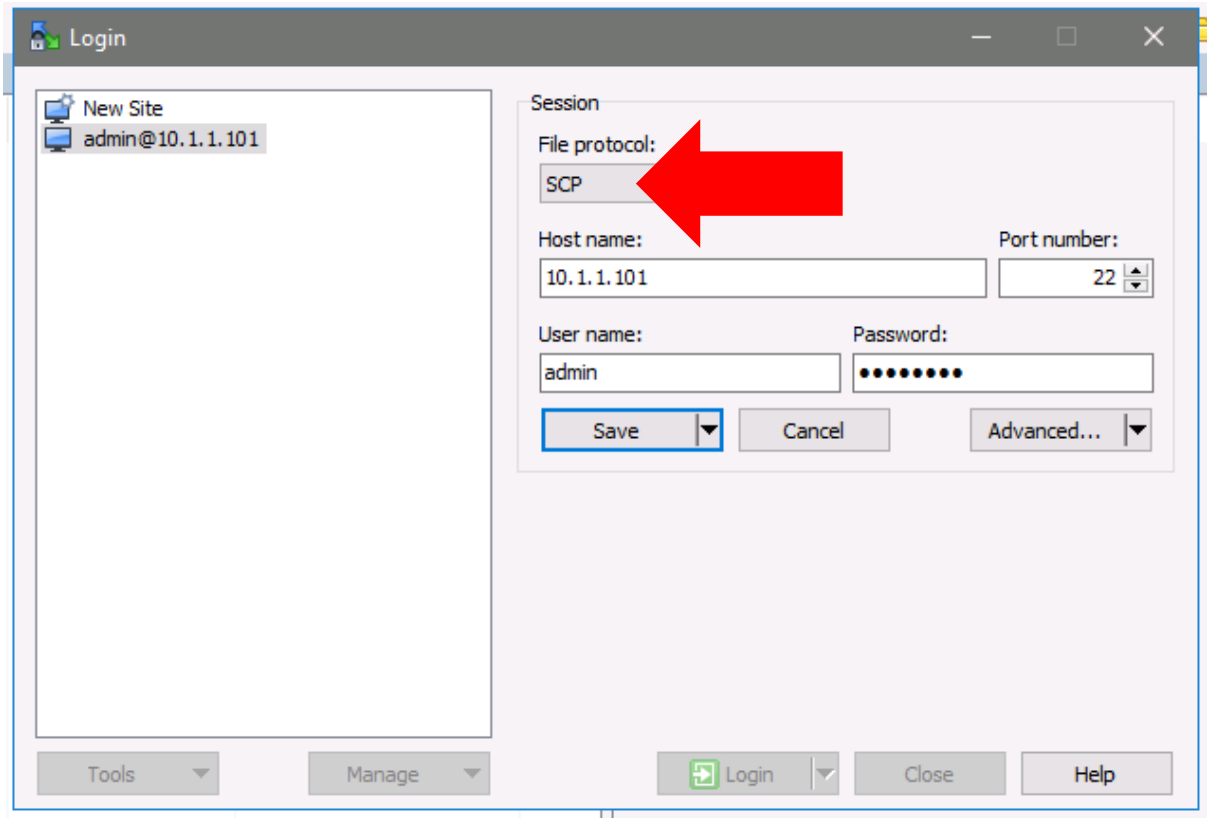
Lab 6

Lab start page: 401

Lab end page: 431

Page 411:

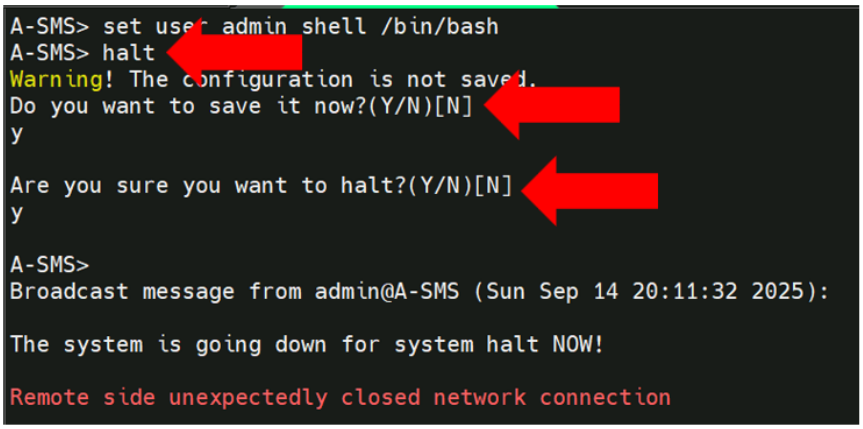
You must use **SCP** as the File protocol in WinSCP:



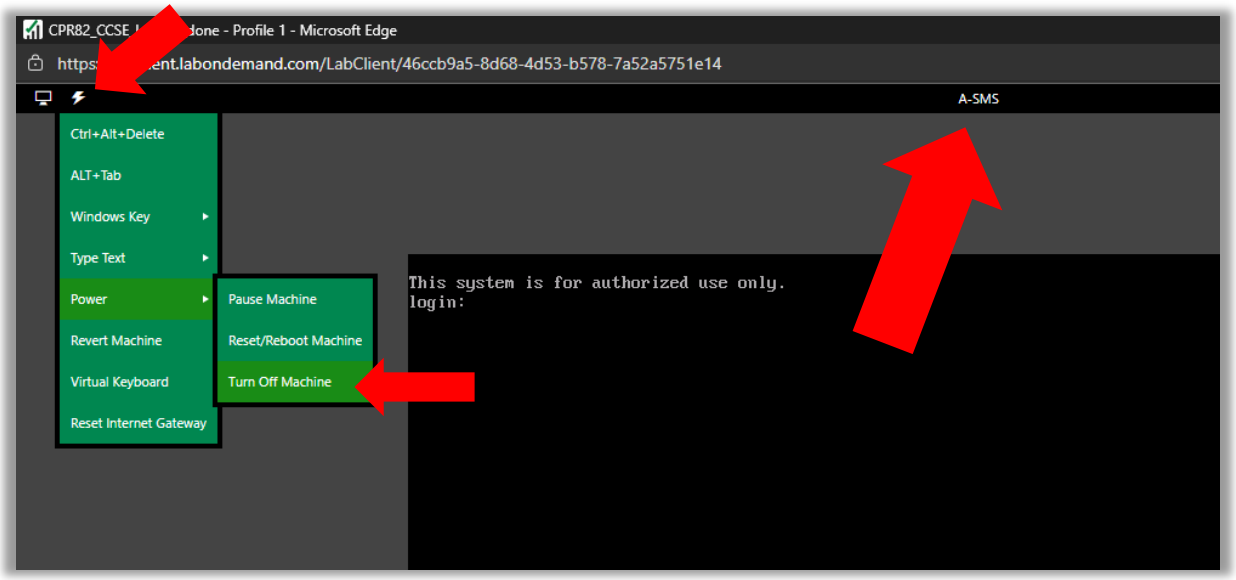
Page 412:

Options for shutting down **A-SMS → Use only one**:

1. Use the **halt** command in an SSH session to shut down A-SMS



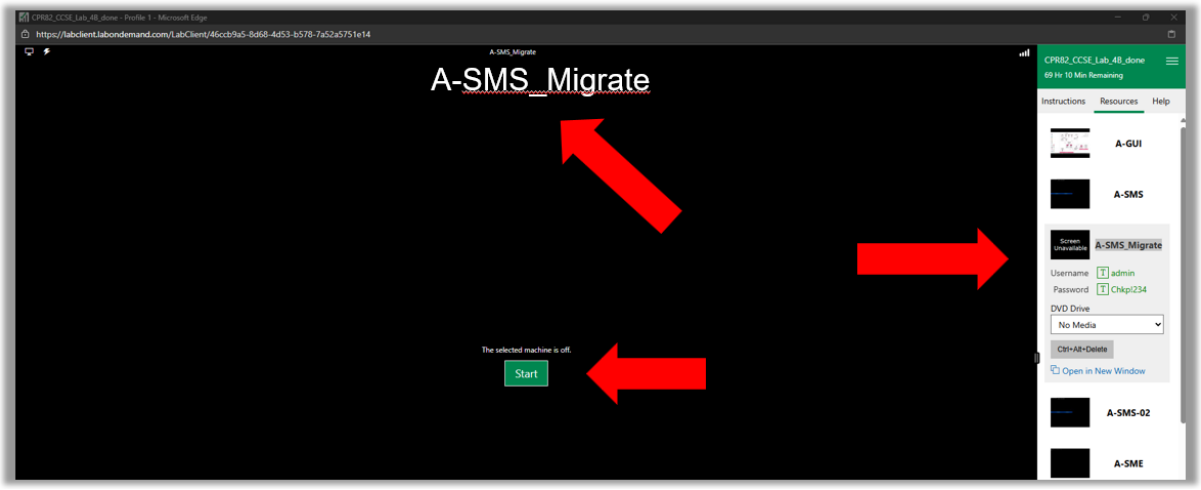
2. Use the Skillable **Commands** drop-down menu > **Power** > **Turn Off Machine**



Page 417:

To start/boot the **A-SMS (Migrate)** VM (**A-SMS_Migrate** in Skillable):

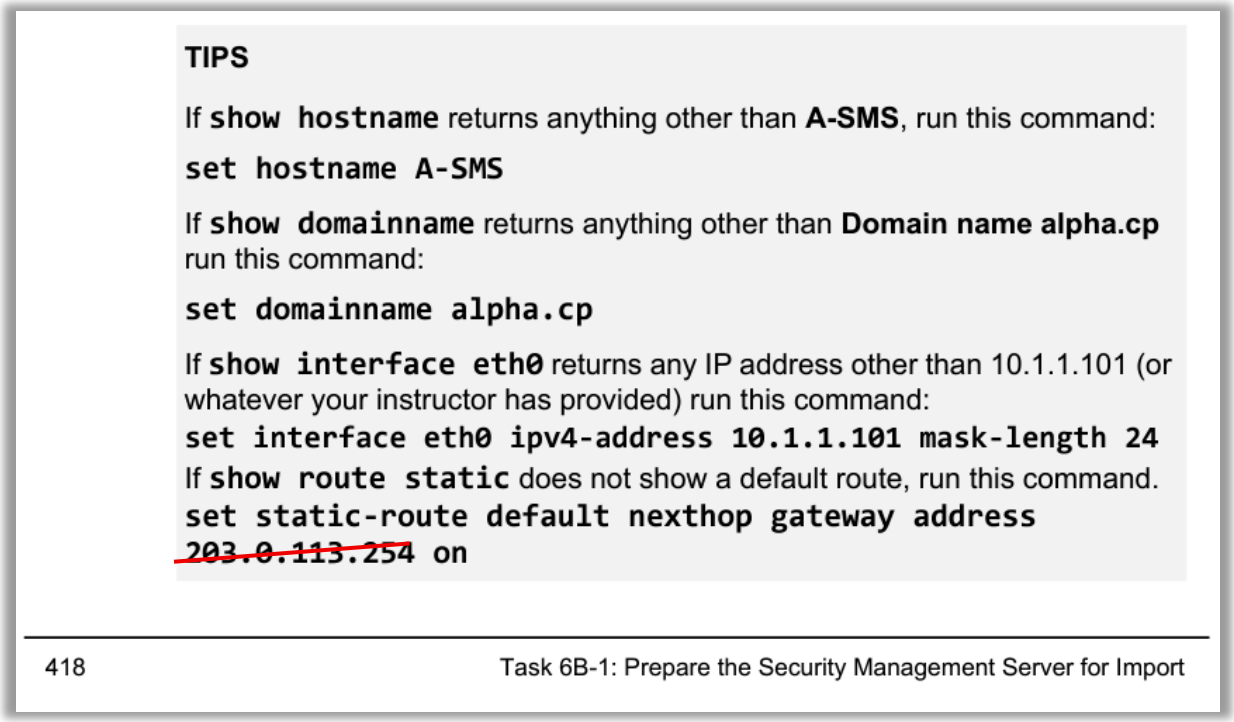
Browse to the **A-SMS_Migrate** VM and click **Start**.



Page 418 – TIPS box:

The set static-route command is wrong.

It is **10.1.1.1** and not 203.0.113.254



TIPS

If **show hostname** returns anything other than **A-SMS**, run this command:

```
set hostname A-SMS
```

If **show domainname** returns anything other than **Domain name alpha.cp** run this command:

```
set domainname alpha.cp
```

If **show interface eth0** returns any IP address other than 10.1.1.101 (or whatever your instructor has provided) run this command:

```
set interface eth0 ipv4-address 10.1.1.101 mask-length 24
```

If **show route static** does not show a default route, run this command.

```
set static-route default nexthop gateway address
203.0.113.254 on
```

418 Task 6B-1: Prepare the Security Management Server for Import

Page 420:

The **api status** command will show FAILED

Please continue and ignore the message.



You need to set the time zone for your new SMS machine.

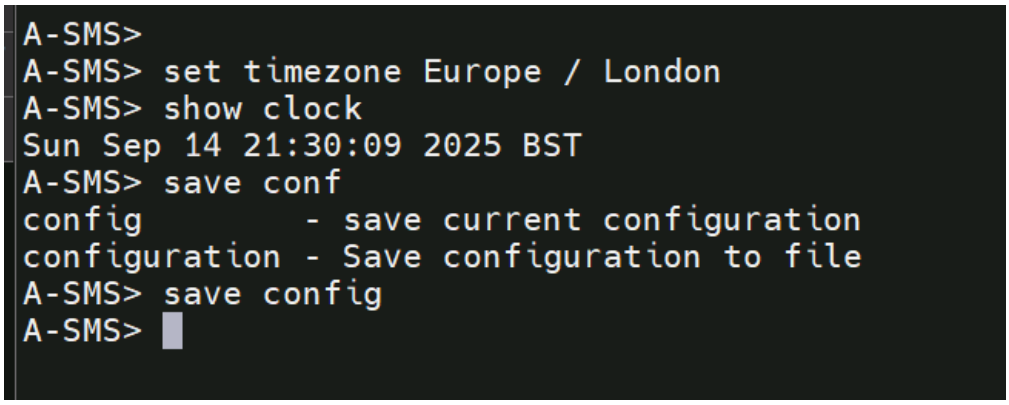
- Use set timezone <TAB Key>

Example:

Setting the time zone to Europe / London:

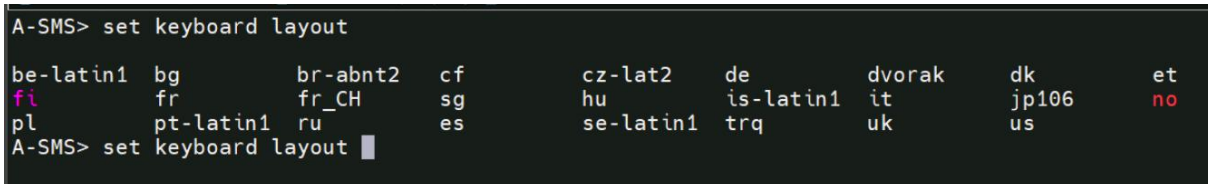
CLISH> **set timezone Europe / London**

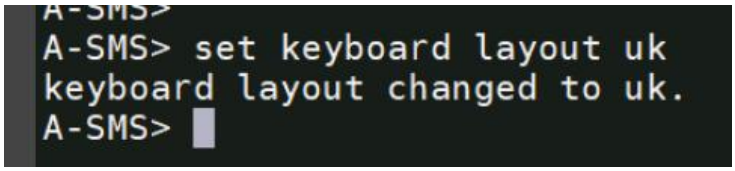
CLISH> **save config**



Setting the **keyboard layout**

CLISH> **set keyboard layout** <TAB>





Example: **uk** for UK keyboard

Page 425:

A-SMS (Migrate) has no valid license.

We need to fix that before continuing. See next page for details.



Page 425:

A-SMS (Migrate) has no valid license (because it was pre-installed).

We need to fix that before continuing.

Required steps (using PowerShell on A-GUI)

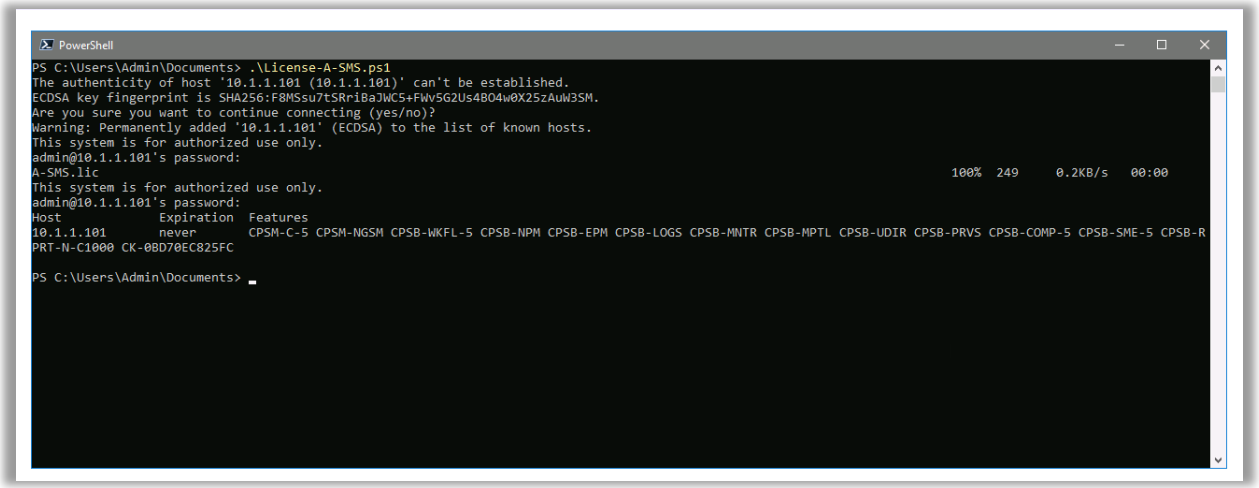
1. scp **A-SMS.lic** file from **A-GUI** to **A-SMS**
2. cplic put -l **A-SMS.lic** ← Run that on the command line of A-SMS

Procedure:

Open PowerShell on A-GUI

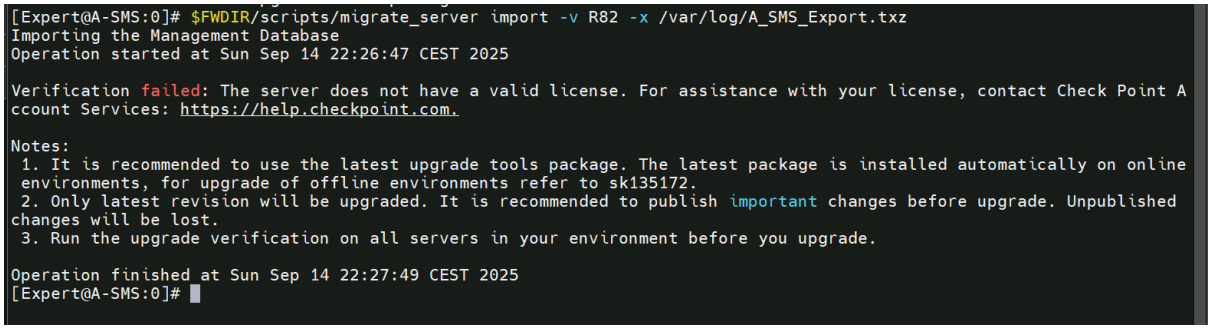
scp "C:\Users\Admin\Desktop\Check Point Tools\Licenses\A-SMS.lic“ admin@10.1.1.101:/home/admin/

ssh admin@10.1.1.101 "cplic put -l A-SMS.lic"



Note:

Import fails with no valid license if licenses are not applied as described above .



Lab 7

Lab start page: **447**

Lab end page: **526**

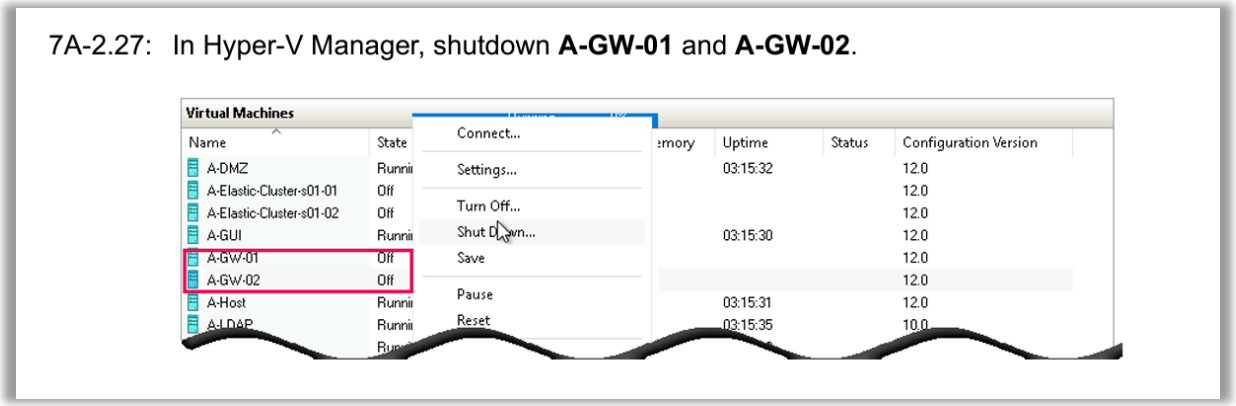
Option to use the Management API in this lab exercise:

During this lab the lab instructions will get you to remove the dependencies on A-GW-Cluster so that it can be deleted. That is using **Where Used** to locate and remove the cluster object from configuration before it can be deleted.

At the bottom of this document are the commands that you can try (optional) instead of using the SmartConsole.

Reminder:

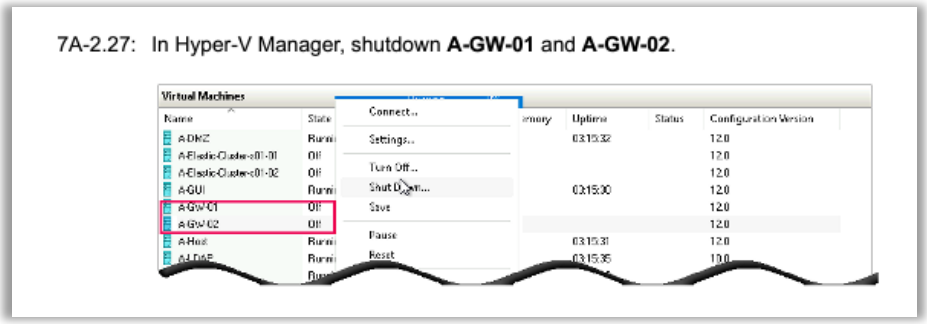
We do not have the Hyper-V Manager in Skillable.



Page 472 – 2.27:

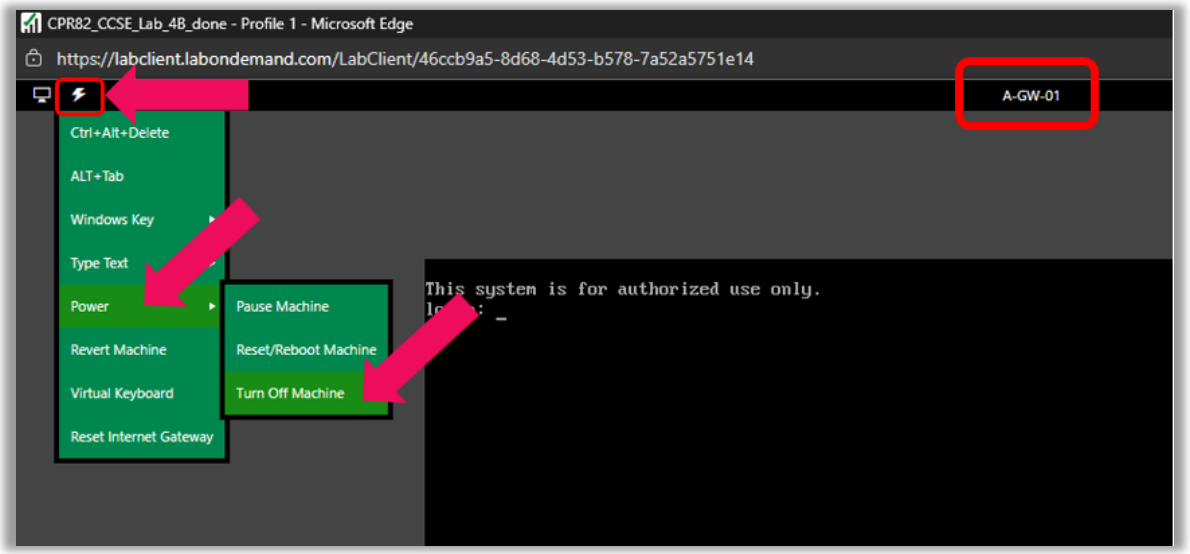
In the Skillable labs there is no Hyper-V Manager access.

Ignore this step and follow the instructions below instead.



To shut down A-GW-01 and A-GW-02 you have two options. Please use one of these options to shut the gateways down.

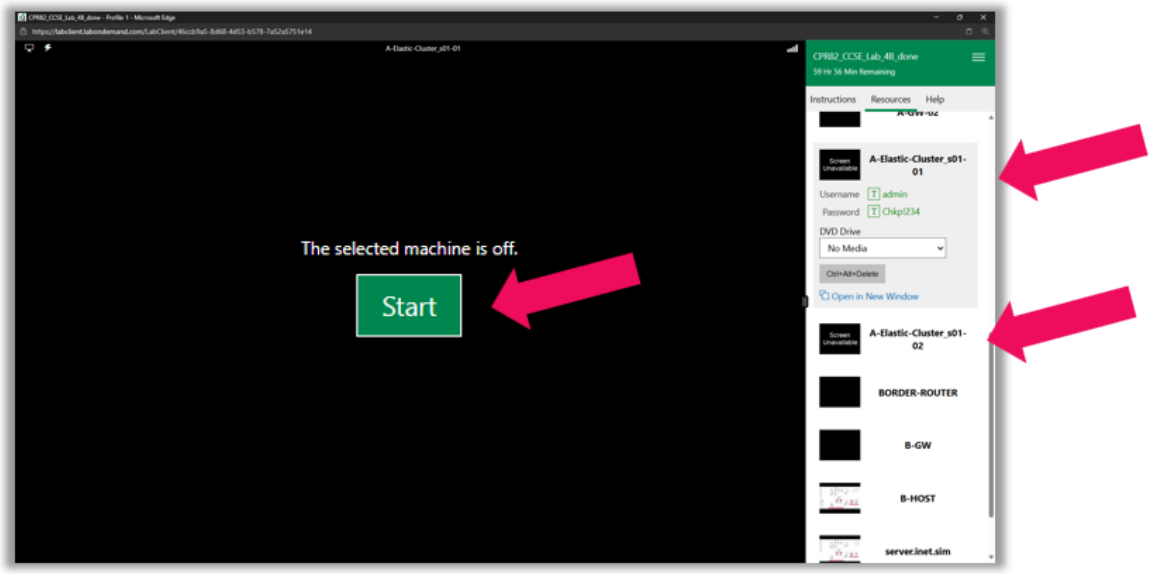
1. SSH to A-GW-01 and A-GW-02 and run the **halt** command.
2. Use the Commands drop-down menu to power off the machines. See screenshot below.



Page 473:

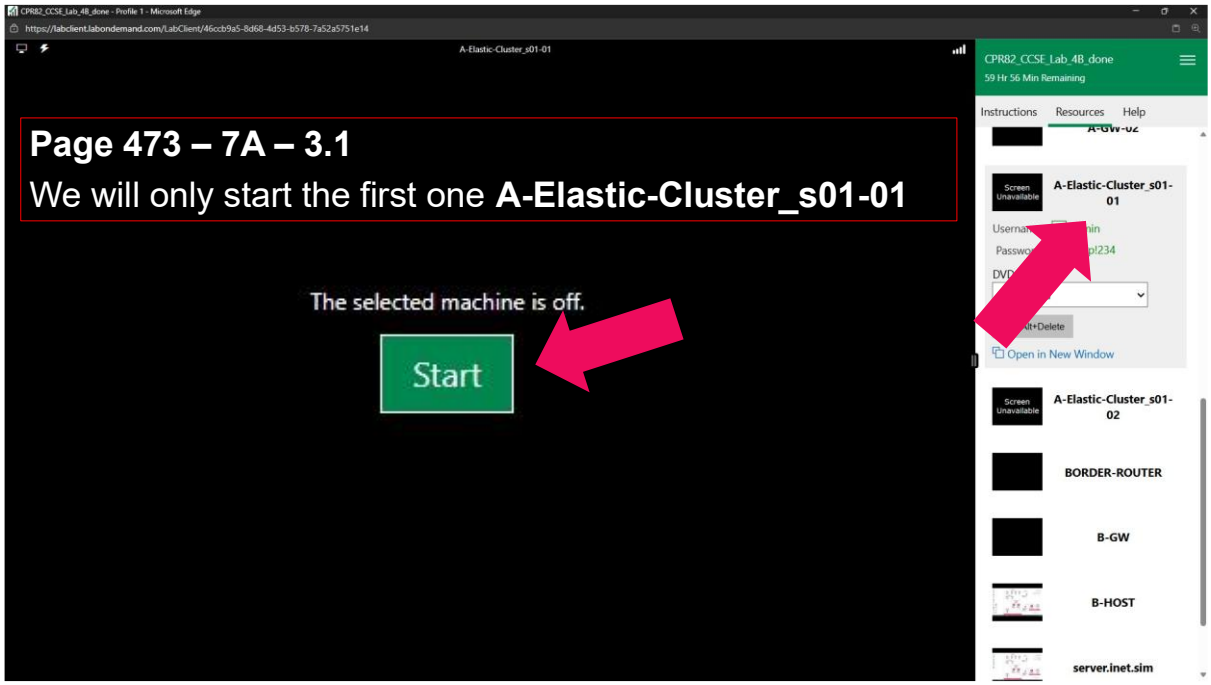
We do not have the Hyper-V Manager in Skillable.

We need to start the ElasticXL gateways using the Labclient interface.



Page 473 – 7A – 3.1

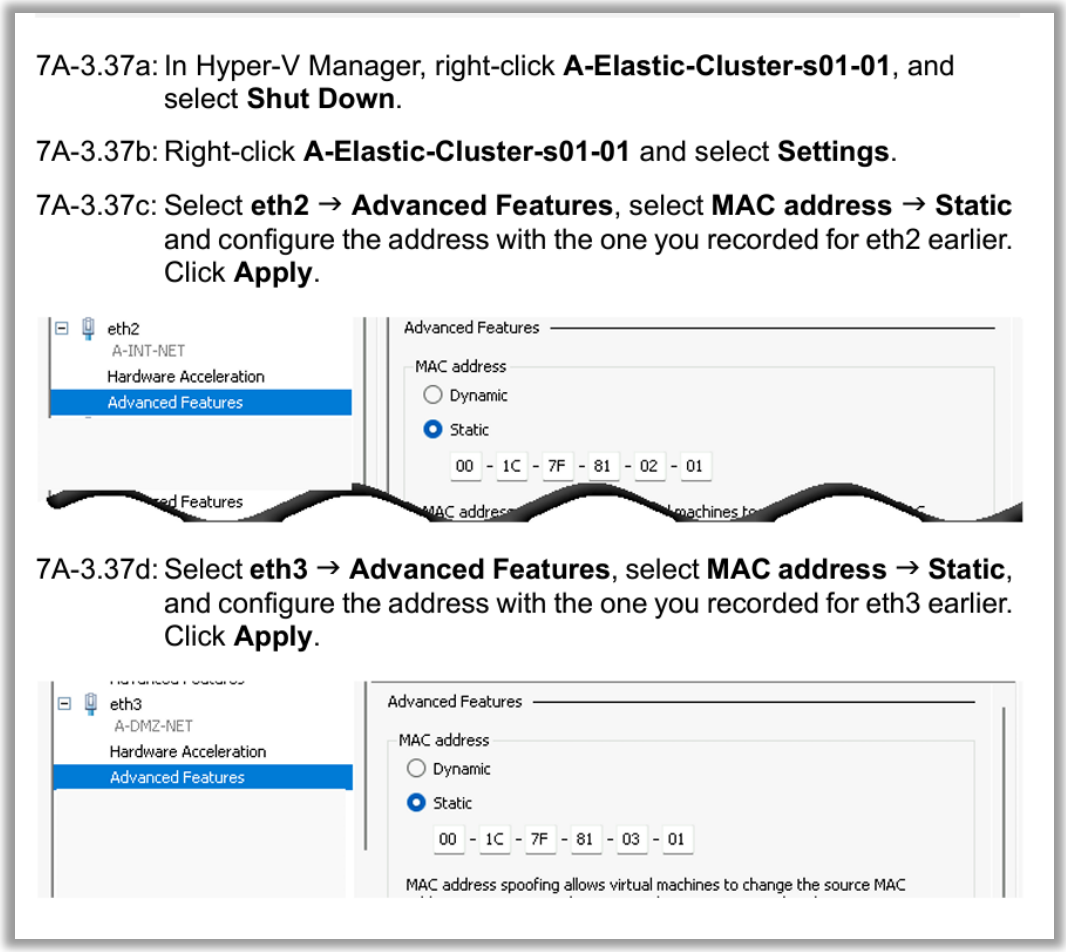
We will only start the first one now: **A-Elastic-Cluster_s01-01**



Page 491 & 492

Ignore the Hyper-V Manager steps.

Jump to Page 493.



Page 509 & 510

The keyboard mapping of **A-Elastic-Cluster_s01-02** is German.

To change that use the CLISH set keyboard layout command.

Example:

set keyboard layout uk

save config

Page 511:

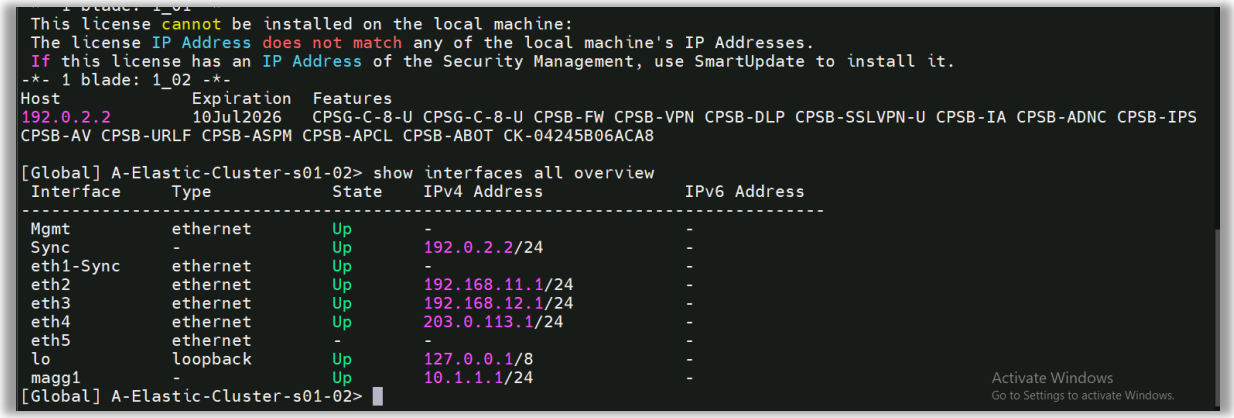
7A – 4-7: The Gaia portal is on port 4434

https://10.1.1.1:4434

Page 513

Ignore the license error and continue.

Note: You can run **cplic print** to check the two nodes' licenses



Note: You can run **cplic print** to check the two nodes' licenses

Page 514

Ignore all of the instruction about Hyper-V.

IMPORTANT

This procedure is required for the Hyper-V ElasticXL to function. This procedure is not supported outside of this lab environment.

7A-4.14c: Run **`show interface eth2`** → record the MAC address for **1_02**.

7A-4.14d: Run **`show interface eth3`** →record the MAC address for **1_02**.

7A-4.14e: Run **`show interface eth4`** → record the MAC address for **1_02**.

7A-4.14f: Run **`exit`** to leave member 2.

7A-4.14g: In Hyper-V Manager, right-click **A-Elastic-Cluster-s01-02** and select **Shut Down**.

7A-4.14h: Right-click **A-Elastic-Cluster-s01-02** and select **Settings**.

7A-4.14i: Select **eth2** → **Advanced Features** →**MAC address** → **Static**.

7A-4.14j: Configure the address with the one you recorded previously for eth2 and click **Apply**.

7A-4.14k: Select **eth3** → **Advanced Features** →**MAC address** → **Static**.

7A-4.14l: Configure the address with the one you recorded previously for eth3 and click **Apply**.

7A-4.14m:Select **eth4** → **Advanced Features** →**MAC address** → **Static**.

7A-4.14n: Configure the address with the one you recorded previously for eth4 and click **Apply**.

7A-4.14o: Click **OK** → right-click **A-Elastic-Cluster-s01-02** →select **Start**.

Lab 7A - Page 453

Removing dependencies on A-GW-Cluster using the Management API (CLI)

These API commands will remove the dependencies on the A-GW-Cluster object so that it can be deleted.

This procedure is not in the lab. The lab follows a different order.

In SmartConsole click the Command Line button in the lower-left corner of the SmartConsole.

These commands can be pasted into the command window (API) that appears.

Reference ---> Lab 7A - Page 453 and 471 --> these commands do not exactly same steps but the procedure is valid.

set network name "A-DMZ-NET" nat-settings.auto-rule false nat-settings.install-on "All"

set network name "A-INT-NET" nat-settings.auto-rule false nat-settings.install-on "All"

set network name "A-MGMT-NET" nat-settings.auto-rule false nat-settings.install-on "All"

set host name A-LDAP nat-settings.auto-rule false nat-settings.install-on "All"

set checkpoint-host name A-SMS nat-settings.auto-rule false nat-settings.install-on "All"

set checkpoint-host name A-SMS-02 nat-settings.auto-rule false nat-settings.install-on "All"

set checkpoint-host name A-SME nat-settings.auto-rule false nat-settings.install-on "All"

set package name "Alpha-Standard" installation-targets all

set vpn-community-meshed name Alpha2BravoVPN gateways.remove A-GW-Cluster

set access-rule layer "Web Content Layer" name "Marketing Access rule" install-on "Policy Targets"

set access-rule layer "Alpha-Standard Network" name "Security Gateway Access rule" destination.remove A-GW-Cluster

set access-rule layer "Alpha-Standard Network" name "Stealth rule" destination.add B-GW

set access-rule layer "Alpha-Standard Network" name "Stealth rule" destination.remove A-GW-Cluster

set nat-rule package "Alpha-Standard" name "DMZ NAT rule" install-on.remove "A-GW-Cluster"

set nat-rule package "Bravo-Standard" name "DMZ NAT rule" install-on.remove "A-GW-Cluster"

